# Decays and LFV Higgs rates in the soft-broken $S_3$ 3HDM: an exact lepton fit at any vacuum, and LFV as pure non-vacuum admixture

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**Abstract**

I have computed the LFV Higgs rates, $h \to \gamma\gamma$ and the heavy-state decays $A \to Zh$, $H^\pm \to W^\pm h$ at all 2021 viable points of the soft-broken scan. In this vacuum the CP-even mixing is a full $3 \times 3$, so the single angle $\delta$ of the exact-$S_3$ analysis no longer applies. Every coupling is computed numerically per point and validated against `feynlag`'s own extractor at a non-aligned point, against the exact-$S_3$ limit, and against the $S_3$ image vacuum. Three results stand out. First, the charged-lepton fit is exact at any vacuum: the mass matrix is orthogonally an arrowhead, so the fit is one cubic per sign pattern. Away from $\varphi = \pi/3$ the draft-type solution survives only for $|\mu_3^\ell| \approx m_\tau$. Second, the vacuum direction's lepton coupling is $\mathrm{diag}(m_\ell)/v$ at every vacuum, so the SM-like state's flavour violation is carried entirely by its $1 - \kappa_V^2$ admixture. Third, all three LFV channels open up, with $h \to e\mu$ the binding one, yet only 5 of 2021 scalar points are excluded for every lepton configuration.

## 1 Summary

Everything below comes from `research/thdm_s3/06_soft_decays.ipynb`, with the numerics in `soft_decays.py` and `lfv.py`.

1. **Validated couplings** (§2): numeric $hH^+H^-$ and VSS couplings agree with the extractor, the exact limit and the $S_3$ image.
2. **The lepton fit off the alignment** is an exact cubic, and soft breaking pins $\mu_3^\ell$ near $\pm m_\tau$ (§3).
3. **SM-like LFV is pure admixture**, and $e\mu$ is the binding channel (§4).
4. $h \to \gamma\gamma$ **and VSS** (§5).

Doing this surfaced an error in the exact-vacuum analysis: its lepton Yukawa split had $G_2 = 0$, which put the gauge-phobic $h_0$'s couplings in the wrong place. That is corrected in the companion scalar-decays report. With the fix, $h_0$ couples only through the electron, exactly as our draft prints.

## 2 Validation

The soft terms are quadratic, so a cubic coupling comes from the quartics alone. The $h_k H_a^+ H_b^-$ couplings are the lambdified tensor $\partial^3 V / \partial r \, \partial c \, \partial \bar{c}$ contracted with each point's rotations. Every

doublet has the same gauge charges, so the VSS couplings are pure geometry:

$$g_{ZA_a h_k} = \frac{g}{2c_W}\,(R_{\rm odd}^{\mathsf T} R_{\rm even})_{ak}, \qquad g_{W H_a^{\pm} h_k} = \frac{g}{2}\,(R_C^{\mathsf T} R_{\rm even})_{ak}.$$

| check | against | agreement |
|---|---|---|
| 27 $h_k H_a^+ H_b^-$ couplings, soft point | `feynlag` extractor | $2 \times 10^{-13}$ GeV |
| 9 $ZAh$ + 9 $WH^{\pm}h$ couplings | `feynlag` extractor | $10^{-17}$ |
| $\Gamma(A \to Zh)$ closed form | decay engine $\lvert M\rvert^2$ | $10^{-10}$ relative |
| $\varphi = \pi/3$, zero soft terms | corrected notebook 03 | gauge-phobic state, trilinears, LFV |
| $\varphi \to 2\pi/3 - \varphi$ (40 points) | itself | $10^{-12}$ |

## 3 The charged-lepton fit off the alignment

In feynlag's basis the doublet block of $M_\ell$ is $a\not{V} + b\,v_{12}\,\mathrm{refl}(\varphi)$, whose eigenvectors $e_\pm$ depend on $\varphi$ only. The third column projects on them as $D(\cos\frac{3\varphi}{2}, \sin\frac{3\varphi}{2})$, so

$$M_\ell \simeq \begin{pmatrix} A & 0 & D\cos\chi \\ 0 & B & D\sin\chi \\ D\cos\chi & D\sin\chi & \mu_3 \end{pmatrix}, \qquad \chi = \tfrac{3\varphi}{2}.$$

Matching the characteristic polynomial to $\prod_i(\lambda - s_i m_i)$ leaves one cubic in $A - B$ per sign pattern. Every branch is verified against the lepton masses. Random-start least squares never finds a solution the cubic misses (377 converged fits). At $\varphi = \pi/3$, $\cos\chi = 0$ and the electron decouples, which recovers the draft. Existence depends on $\cos 3\varphi$ only.

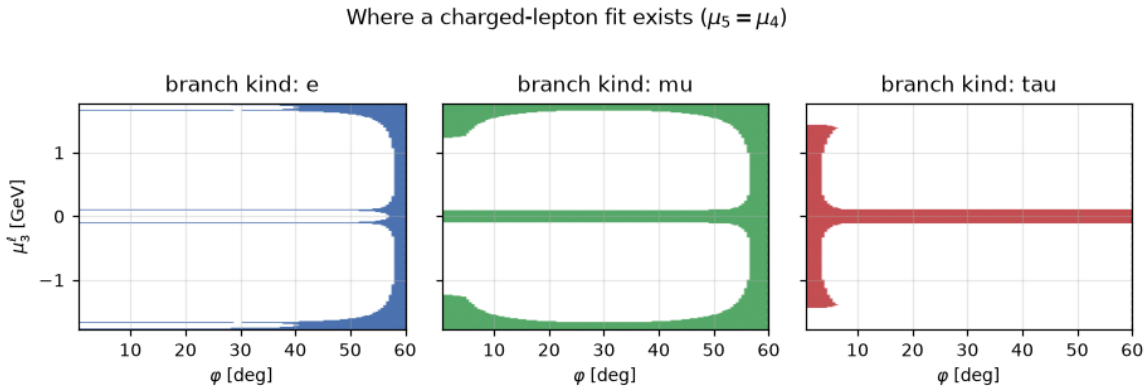



Figure 1: Where a lepton fit exists, by branch kind (which lepton the $e_+$ entry $A$ grows into; "e" is the draft's branch). At $\varphi = 60°$ the whole $\mu_3$ axis is allowed. Away from it the draft-type branch survives on about 5% of the axis, near $\pm m_\tau$.

So soft breaking turns the exact model's free dial $\mu_3^\ell \in (m_\mu, m_\tau)$ into a near-prediction: the third generation is essentially the $S_3$ singlet.

## 4 LFV rates

The vacuum direction $\hat{v}$ couples to leptons through $\sum_k v_k G_k / v = M_\ell / v$, which is diagonal in the mass basis at *any* vacuum (checked to $10^{-17}$ at 203 points). The SM-like state's flavour violation

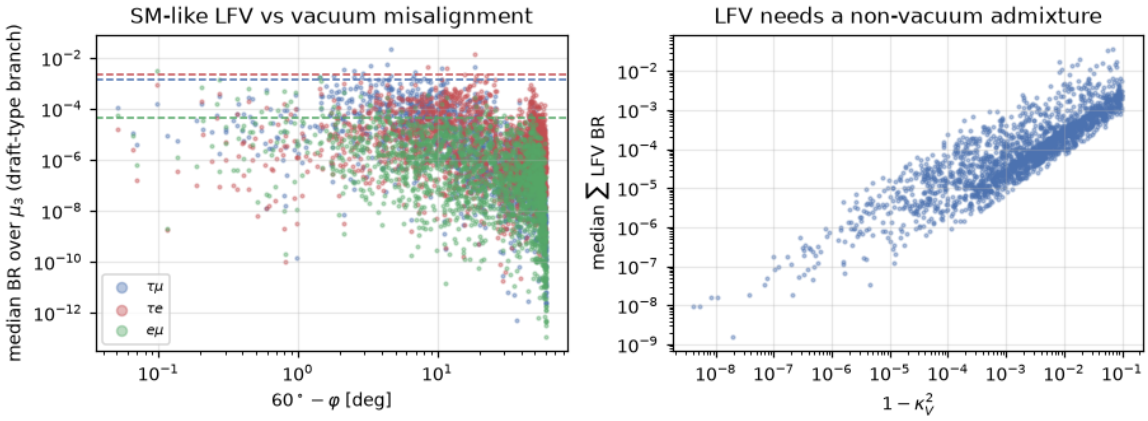



Figure 2: Left: SM-like LFV rates (median over $\mu_3$, draft-type branch) against the vacuum misalignment, with the CMS limits dashed. Right: the total LFV rate against the non-vacuum admixture $1 - \kappa_V^2$; the correlation is 0.88 in log.

therefore comes entirely from its admixture of the other two directions, of weight $1 - \kappa_V^2$ (median $2.4 \times 10^{-3}$). This is the alignment limit, and it holds whatever the size of the soft breaking (Fig. 2).

Over every (point, $\mu_3$, branch) entry, with $\Gamma_h = 4.1$ MeV, the SM prediction for the total width [5] (the standard denominator of LFV searches):

| SM-like channel | median $\mathcal{B}$ | entries over the limit | limit (95% CL) |
|---|---|---|---|
| $\tau\mu$ | $9.4 \times 10^{-6}$ | 2.9% | $1.5 \times 10^{-3}$ [2] |
| $\tau e$ | $9.3 \times 10^{-6}$ | 1.9% | $2.2 \times 10^{-3}$ [2] |
| $e\mu$ | $2.6 \times 10^{-6}$ | **18.3%** | $4.4 \times 10^{-5}$ [3] |

$e\mu$ binds hardest. Still, only **5 of 2021** scalar points have no lepton configuration under all three limits. As at the exact vacuum, LFV data bound the lepton parameters rather than the scalar sector.

## 5 $h \to \gamma\gamma$ and the heavy states

$R_{\gamma\gamma} = \Gamma/\Gamma_{\rm SM}$ uses both charged Higgses in the loop, with coefficient $g_{hH^+H^-}v/(2m_{H^\pm}^2)$ [4], and the top coupling *assumed* equal to $\kappa_V$. The median is 0.97, and 90% of points lie in $[0.85, 1.00]$. The 48 points below 0.8 all have a light $H^\pm$ (below 310 GeV) interfering destructively with the $W$. The charged loop decouples to $\kappa_V^2$ above 1.5 TeV.

$A_{1,2} \to Zh$ and $H_{1,2}^\pm \to W^\pm h$ are open at nearly every point, with median widths of 20–80 MeV. Their couplings to the SM-like state obey $\sum_a |O_{ak}|^2 = 1 - \kappa_V^2$ exactly, so they vanish in the alignment limit as $hVV$ saturates.

## 6 Open questions

1. **The quark sector per point.** The heavy-state total widths and the top coupling in $R_{\gamma\gamma}$ both need it. The soft quark fit exists at one vacuum only.

2. $R_{\gamma\gamma}$ **as a cut.** It needs the production side and a cited signal strength.

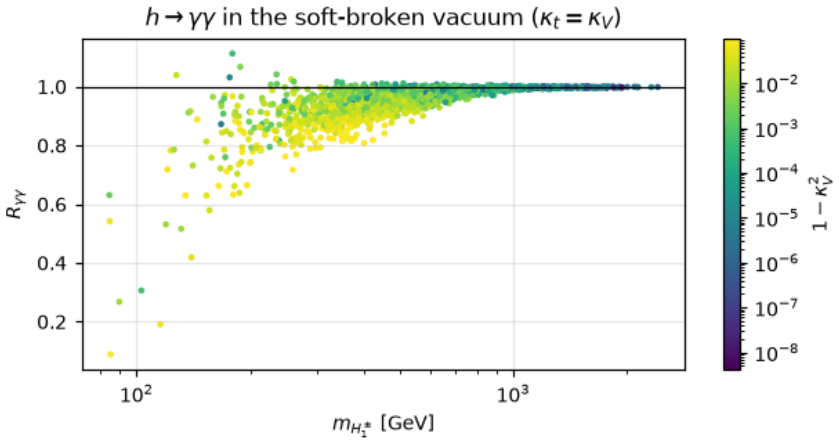



Figure 3: $R_{\gamma\gamma}$ against the lighter charged-Higgs mass, coloured by the non-vacuum admixture.

3. $\mu \to e\gamma$**.** Done in the companion $\mu \to e\gamma$ report (notebook 07): it is forbidden at the exact vacuum, and in the soft vacuum MEG II keeps 610 of the 2021 points.

4. **The $\mu_3$ prior.** The quoted fractions are over a uniform $\mu_3$ grid and all branches; the 5-point exclusion count is prior-independent.

## References


[1] M. Zeleny-Mora, M. Mondragón and T. A. Valencia-Pérez, *Exploring LFV Higgs decays in the Three Higgs Doublet Model*, draft (`paper_lfvhd/LFVHD_3HDMS3.tex`).

[2] CMS Collaboration, *Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in pp collisions at $\sqrt{s} = 13$ TeV*, Phys. Rev. D **104**, 032013 (2021), arXiv:2105.03007.

[3] CMS Collaboration, *Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV*, Phys. Rev. D **108**, 072004 (2023), arXiv:2305.18106, doi:10.1103/PhysRevD.108.072004.

[4] M. Chakraborti, D. Das, M. Levy, S. Mukherjee and I. Saha, *Prospects of light charged scalars in a three Higgs doublet model with $Z_3$ symmetry*, arXiv:2104.08146.

[5] LHC Higgs Cross Section Working Group (D. de Florian *et al.*), *Handbook of LHC Higgs Cross Sections: 4. Deciphering the Nature of the Higgs Sector*, CERN Yellow Rep. Monogr. **2** (2017), arXiv:1610.07922.